

Avi Kadria

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Profile

PhD candidate in Computer Science at Bar-Ilan University, researching **distance-related graph algorithms**, with papers at **SODA, ICALP, ESA, ISAAC, and DISC**. **Completed my BSc during high school**, followed by an MSc with two SODA papers. Former IDF officer with 5+ years in Unit 8200, where I **built a new cyber R&D team from the ground up**, led critical projects, and recruited for its elite tracks; now researching full-time.

Education

Bar-Ilan University

Ramat Gan, Israel

PhD in Computer Science

Oct 2023 - Now

- Researching distance-related graph problems, including **distance oracles, girth, diameter, routing, and fine-grained complexity**.
- Published 7 papers in graph algorithms (listed below and at avika81.github.io/papers).
- Recipient of the **Clore Scholarship** for exceptional PhD students (2025) and the **President Scholarship** (2024).

Bar-Ilan University

Ramat Gan, Israel

MSc in Computer Science

Oct 2017 - Mar 2023

- Researched **girth approximations**, published and presented two papers at **SODA 2022** and **SODA 2023**.
- Collaborated with **leading global experts** in distance-related problems.
- Recipient of the **Nadav Scholarship** for exceptional research, as the sole master's student selected (2022).
- Started during the third year of the BSc, and was a full-time officer at the IDF for most of the degree.

Bar-Ilan University

Ramat Gan, Israel

BSc in Computer Science and Pure Mathematics

Jul 2015 - Aug 2018

- Graduated with honors at the age of 17.

Publications

Improved Approximation Algorithms for n -Pairs Shortest Paths

Avi Kadria, Liam Roditty, Virginia Vassilevska Williams

34th Annual European Symposium on Algorithms, ESA 2026, L'Aquila, Italy, 2026

Tighter Bounds for Weighted and Unweighted Shortest Cycle Approximation

Avi Kadria, Liam Roditty, Virginia Vassilevska Williams

34th Annual European Symposium on Algorithms, ESA 2026, L'Aquila, Italy, 2026

Faster Algorithms for $(2k-1)$ -Stretch Distance Oracles

Avi Kadria, Liam Roditty

53rd International Colloquium on Automata, Languages, and Programming, ICALP 2026, Egham, United Kingdom, 2026

New Approximate Distance Oracles and Their Applications - **best paper award winner**

Avi Kadria, Liam Roditty

36th International Symposium on Algorithms and Computation, ISAAC 2025, Tainan, Taiwan, 2025

Compact Routing Schemes in Undirected and Directed Graphs

Avi Kadria, Liam Roditty

39th International Symposium on Distributed Computing, DISC 2025, Berlin, Germany, 2025

Improved girth approximation in weighted undirected graphs - **Accepted to SICOMP'26**

Avi Kadria, Liam Roditty, Aaron Sidford, Virginia Vassilevska Williams, Uri Zwick

Proceedings of the 2023 ACM-SIAM Symposium on Discrete Algorithms, SODA 2023, Florence, Italy, 2023

Algorithmic trade-offs for girth approximation in undirected graphs

Avi Kadria, Liam Roditty, Aaron Sidford, Virginia Vassilevska Williams, Uri Zwick

Proceedings of the 2022 ACM-SIAM Symposium on Discrete Algorithms, SODA 2022, Virtual Conference / Alexandria, VA, USA, 2022

Awards

2026 **Rector's Award for Outstanding Students**, Computer Science

BIU

2026 **2nd place, Feder Family Award for Best Student Work in Communications**, Computer Science

Israel

2025 **Best Paper Award, ISAAC 2025**, Computer Science

Taiwan

2025 **1st place, Israeli Bughouse Championship (2018, 2024, 2025)**, Chess

Israel

2022 **3rd place, Bughouse World Championship**, Chess

Germany

2018 **IMO-trial and IOI-trial Olympiad Excellence**, Mathematics and Computer Science

Israel

2011 **Ranked 5th in Israel (Under 14)**, Chess

Israel

Military Service

Unit 8200
Cyber R&D Team Lead

Tel Aviv, Israel
May 2022 - May 2024

- Formed a new team of Cyber R&D developers to solve crucial problems in the core of the unit's operations.
- Multidisciplinary team operating at cyber security and intelligence interface. Worked in **Python, C, ASM, HTML/JS, and SQL**.
- Mentored new team members, built the team's training program, and led numerous cross-team projects.

Unit 8200
Recruitment officer for ARAM, GA'ASH, and ARAZIM

Tel Aviv, Israel
Aug 2021 - May 2022

- Oversaw and redesigned the entire recruitment process for the most elite tracks of Unit 8200, conducting over 500 interviews.

Unit 8200
R&D engineer

Tel Aviv, Israel
Feb 2019 - Aug 2021

- Practical cyber security operations in Windows environments and large networks. Worked mainly in **Python and C++**.
- Developed new systems and optimized existing methods to increase efficiency and achieve outcomes previously unattainable.

Work Experience

D.S.E
Algorithms Engineer

U.S.A
May 2018 - Feb 2019

- Applied linear programming to develop and optimize employee scheduling algorithms, collaborating with multi-country teams.

Skills and Interests

Programming

Python, C, C++, Rust, ASM, (passable in Java, C#, HTML, JS, SQL, MATLAB, etc.)

Miscellaneous

Git, bash, Docker, ~~TeX~~TeX, Hebrew (native), English (fluent), French (passable). Interests: chess, bughouse, reading.